SUPPLEMENTAL MATERIALS

Evaluating Agentic Retrieval-Augmented Generation Architectures for Bridge Engineering and Nondestructive Evaluation

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The following figures provide additional FHWA InfoTechnology chatbot examples that support the manuscript. Supplemental Figures S1–S5 provide representative examples of the system's responses to different bridge engineering and nondestructive evaluation (NDE) queries. Each case illustrates how the proposed retrieval-augmented generation framework retrieves and synthesizes information from the FHWA InfoTechnology corpus for a specific engineering information need. The examples include different query types and NDE applications and are provided to complement the quantitative evaluation presented in the main manuscript by showing representative system outputs and their supporting engineering context.

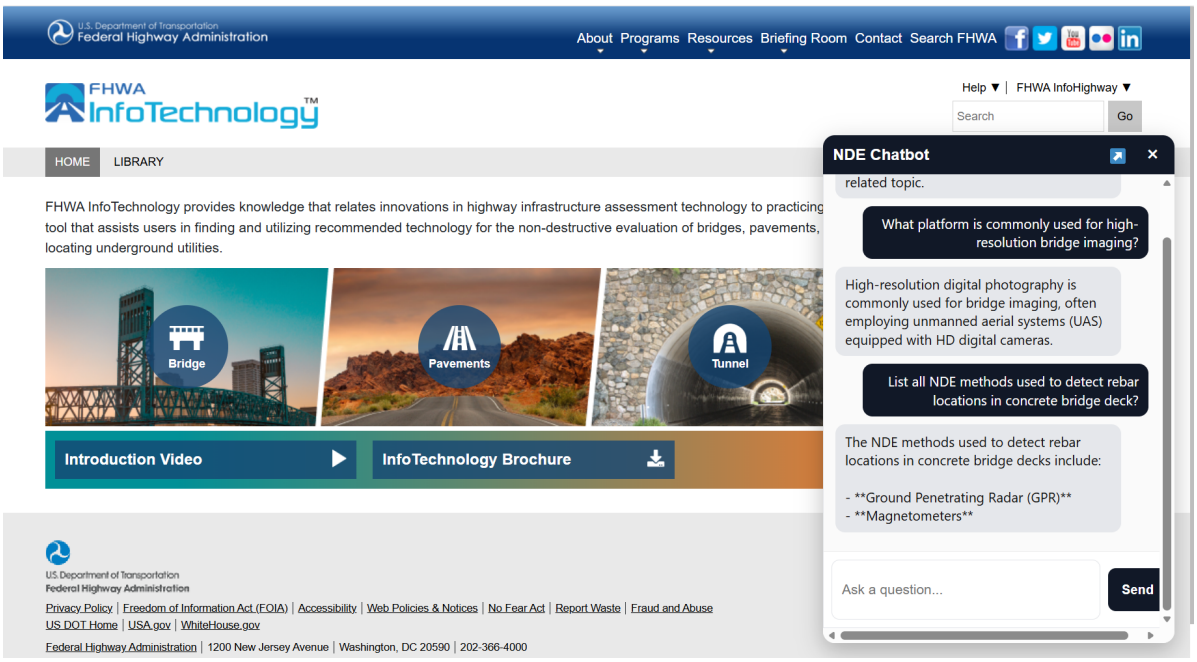


Fig. S1. FHWA InfoTechnology chatbot example showing a grounded response to a bridge-imaging question and a list-type query on rebar-location methods.

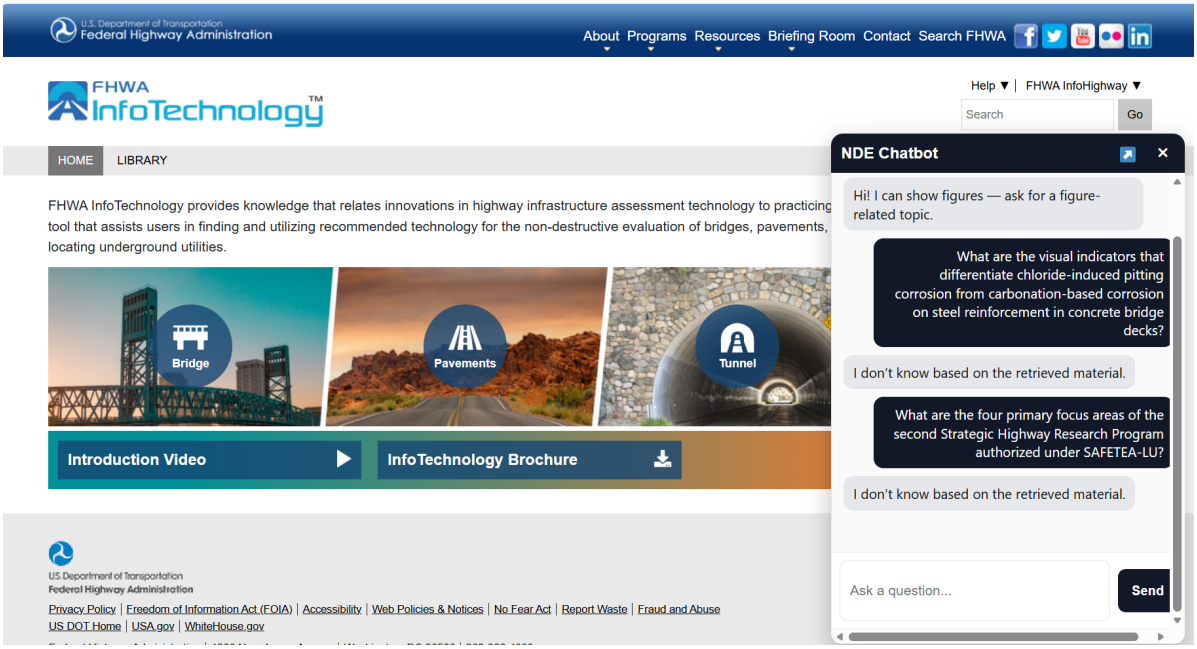


Fig. S2. FHWA InfoTechnology chatbot example showing refusal behavior when the retrieved material does not support an answer.

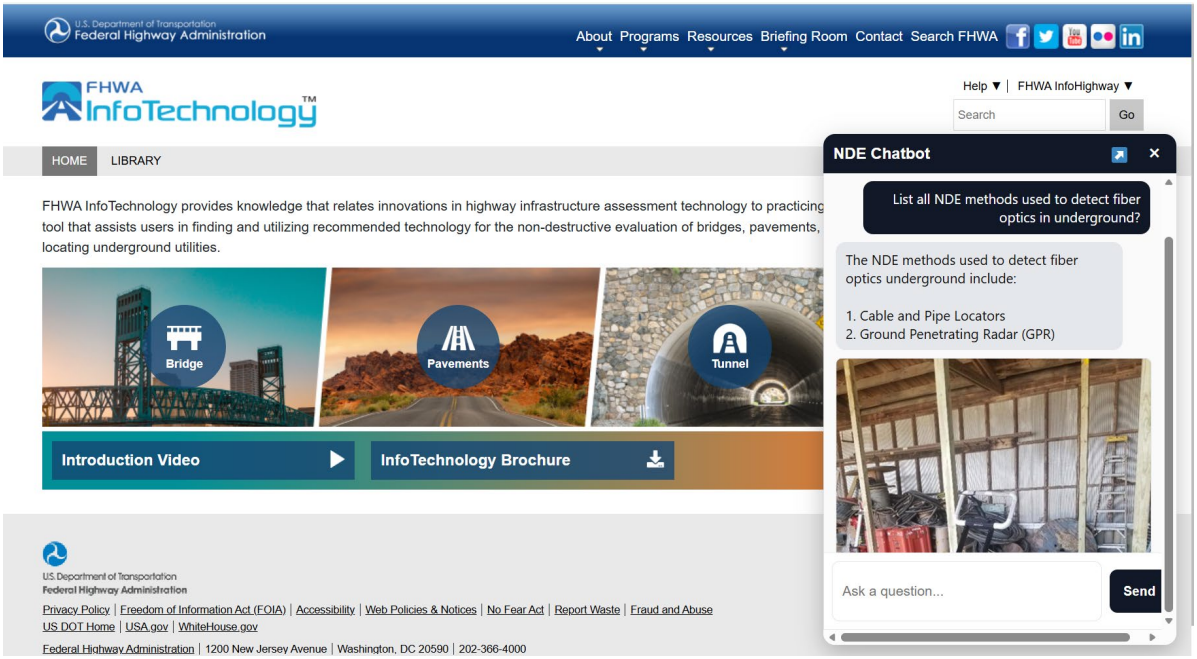


Fig. S3. FHWA InfoTechnology chatbot example showing a list-type response with retrieved visual evidence.



Fig. S4. FHWA InfoTechnology chatbot example showing a list-type response with supporting NDE imagery.

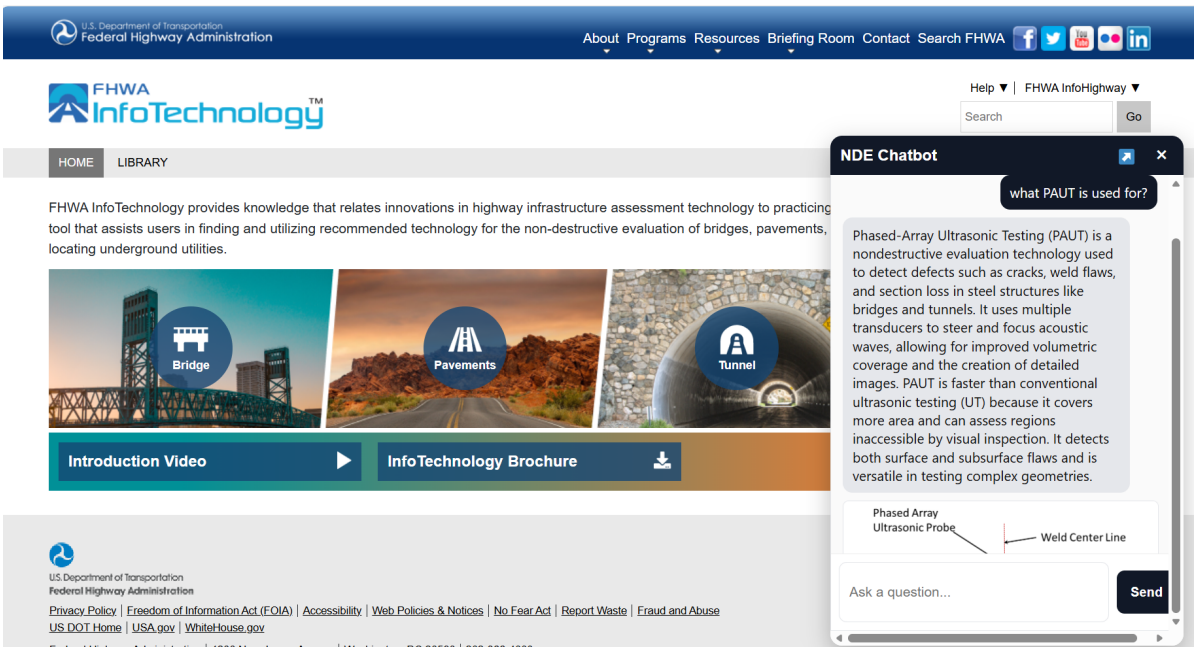


Fig. S5. FHWA InfoTechnology chatbot example showing an explanatory response for phased-array ultrasonic testing with supporting visual material.